

JavaScript Basics — Assignment 04

Operators & Conditional Statements | Beginner Level

Topics Covered: Arithmetic Operators, Assignment Operators, Comparison Operators, if, else if, else, Ternary Operator, switch Statements

Important Instructions

- Read all questions carefully before starting.
- Create a **separate JavaScript file for each question**.
- You may create a separate HTML file for each question where required to run and demonstrate your JavaScript.
- Write clean, well-structured, and properly indented code.
- Use only the JavaScript concepts taught in class.
- Do not use JavaScript frameworks or libraries.
- Test your code in the browser and check the Console where required.
- Students should be able to explain the logic and operators used in their code.

Submission Instructions

- Submit the assignment by providing a **link to your GitHub repository**.
- Keep a **separate file for each question**.
- Push all your final files to GitHub before submitting the repository link.
- Make sure the repository is accessible to the instructor.

Question 1 — JavaScript Operators

Create a JavaScript program that demonstrates your understanding of **Arithmetic, Assignment, and Comparison Operators**.

Tasks:

- Create two number variables, for example **num1** and **num2**.
- Use arithmetic operators to perform addition, subtraction, multiplication, division, modulus, increment, and decrement.
- Create a separate variable and demonstrate assignment operators such as **=**, **+=**, **-=**, ***=**, and **/=**.
- Use comparison operators to compare two values using operators such as **==**, **===**, **!=**, **!==**, **>**, **<**, **>=**, and **<=**.
- Display the results using **console.log()**.

Question 2 — Student Grade Calculator Using if / else if / else

Create a JavaScript program that takes a student's marks and determines the student's grade using **if**, **else if**, and **else**.

Requirements:

- Create a variable named **marks** and assign it a value between 0 and 100.
- Use **if**, **else if**, and **else** statements to determine the grade.
- Use the following grading criteria:

Marks	Grade
90–100	A
80–89	B
70–79	C
60–69	D
Below 60	F

- Display the student's marks and grade using **console.log()**.
- Test your program with at least three different marks.

Question 3 — Login Validation Using if / else

Create a simple login validation program using JavaScript. The program should check whether the entered username and password are correct.

- Create variables for **username** and **password**.
- Set a predefined correct username and password in separate variables.
- Use **if** and **else** to compare the entered credentials with the correct credentials.
- If both are correct, display “**Login Successful**”.
- If either value is incorrect, display “**Invalid Username or Password**”.
- Use comparison operators appropriately.
- Display the result using **alert()** or **console.log()**.

Question 4 — Ternary Operator

Create a JavaScript program that uses the **ternary operator** instead of a traditional if/else statement.

- Create a variable named **age** and assign it a value.
- Use the ternary operator to determine whether the person is eligible to vote.
- If the age is 18 or above, display “**Eligible to Vote**”.
- Otherwise, display “**Not Eligible to Vote**”.
- Display the result using **console.log()**.

Note: The main purpose of this question is to practice the ternary operator syntax: condition ? valueIfTrue : valueIfFalse.

Question 5 — Menu Selection Using switch

Create a JavaScript program that displays a simple menu and uses a **switch statement** to respond to the user's selection.

- Create a variable named **choice** and assign it a number from 1 to 4.
- Use a **switch** statement to handle the choices.

Choice	Option	Output
1	Home	You selected Home
2	About	You selected About

3	Services	You selected Services
4	Contact	You selected Contact

- Add a **default** case for an invalid choice.
- Display the appropriate output using **console.log()** or **alert()**.

Final Submission Checklist

- Separate file created for each question.
- Arithmetic, assignment, and comparison operators demonstrated.
- if / else if / else demonstrated.
- if / else login validation completed.
- Ternary operator demonstrated.
- switch statement with default case completed.
- Code tested in the browser.
- All files pushed to GitHub.
- GitHub repository link is ready to submit.

Happy Coding! ■